Phonon transmission across a loosely bonded crystal/metal interface: derivation and implementation notes

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Abstract

Technical note accompanying the code in this repository. It derives the reflection and transmission amplitudes of plane elastic waves at a planar interface described by tangential and normal spring constants (the loose-bond acoustic mismatch model of Every, Koos and Wolfe [1]), the power coefficients, the flux and Debye averages, and the closed-form limits used to validate the implementation.

This appendix derives the reflection and transmission amplitudes used in the thesis chapter: the field expressions for each wave, the coefficient matrix of the boundary-value problem, the SH channel, the power coefficients, and closed-form checks.

1 Material parameters

Table 1 lists the parameters entering the model. The Lamé parameters for sapphire, Al, Nb and Ta are those of the thesis (Table 1 lists them); the velocities follow from $V_T = \sqrt{\mu/\rho}$ and $V_L = \sqrt{(\lambda + 2\mu)/\rho}$. Cu is not part of that calculation and retains the values of Ref. [1].

Table 1: Isotropic parameters used in the interface model. Densities are assumed bulk values; Cu values are from Ref. [1].

Material	λ [GPa]	μ [GPa]	ρ [kg m^{-3}]	V_L [m/s]	V_T [m/s]
Sapphire	139.8	166.24	3986	10885	6458
Al	58.7	26.2	2700	6415	3115
Nb	145	39.7	8570	5117	2152
Ta	149.3	70.55	16650	4176	2058
Cu	106.0	45.6	8930	4700	2260

2 Geometry, conventions and unknowns

The interface is the plane $z = 0$. The crystal (sapphire, subscript c) occupies $z < 0$ and the metal (subscript m) occupies $z > 0$. The x axis lies in the interface, and all fields are independent of y. The time dependence is $e^{-\mathrm{i}\omega t}$. Each medium is isotropic with density ρ, wave speeds V_L, V_T and Lamé constants

$$\mu = \rho V_T^2, \qquad \lambda = \rho V_L^2 - 2\mu, \qquad \lambda + 2\mu = \rho V_L^2. \tag{1}$$

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Because z points from the crystal into the metal, the incident and transmitted waves travel towards $+z$ and the reflected waves towards $-z$. This is the opposite of the drawing in Fig. 1(a), which has the crystal on top, but the physics is identical.

3 Wave equation and Helmholtz potentials

In each medium

$$\rho\,\ddot{\mathbf{u}} = (\lambda+2\mu)\nabla(\nabla\cdot\mathbf{u}) - \mu\,\nabla\times\nabla\times\mathbf{u}. \tag{2}$$

For motion in the x–z plane write

$$\mathbf{u} = \nabla\phi + \nabla\times(\psi\,\hat{\mathbf{y}}), \qquad \ddot{\phi} = V_L^2\nabla^2\phi, \qquad \ddot{\psi} = V_T^2\nabla^2\psi, \tag{3}$$

which gives $u_x = \partial_x\phi - \partial_z\psi$ and $u_z = \partial_z\phi + \partial_x\psi$. The scalar ϕ carries the longitudinal (L) wave and ψ the in-plane transverse (SV) wave. The component u_y (SH) is decoupled and treated in Sec. 9. The stress is

$$\sigma_{ij} = \lambda\,\delta_{ij}\,\nabla\cdot\mathbf{u} + \mu\,(\partial_i u_j + \partial_j u_i). \tag{4}$$

4 Plane waves, conserved slowness and Snell's law

Take plane waves

$$\phi = A\,e^{\mathrm{i}\omega(px + s\,k_L z - t)}, \qquad \psi = B\,e^{\mathrm{i}\omega(px + s\,k_T z - t)}, \tag{5}$$

where $s = \pm 1$ is the sign of the vertical propagation direction and p is the horizontal slowness. The boundary conditions must hold for every x at $z = 0$, so every wave of the problem has the same p, which is Snell's law:

$$p = \frac{\sin\theta_i}{V_i} = \frac{\sin\theta_j}{V_j} \quad \text{for every wave } j. \tag{6}$$

Substituting into the wave equations gives the vertical slowness

$$k_j(p) = \sqrt{V_j^{-2} - p^2}\,, \qquad \cos\theta_j = V_j\,k_j, \tag{7}$$

for $j \in \{L, T\}$ in either medium. If $p > 1/V_j$, then k_j is purely imaginary. The branch $\operatorname{Im} k_j \geq 0$ is chosen, $k_j = \mathrm{i}\sqrt{p^2 - V_j^{-2}}$, so that the wave decays away from the interface (it decays for $z \to +\infty$ with $s = +1$ and for $z \to -\infty$ with $s = -1$). Such a wave is evanescent and carries no energy.

5 Displacements and stresses of each wave

Differentiating the plane waves gives, for an L wave,

$$u_x = \mathrm{i}\omega p\,A, \qquad u_z = \mathrm{i}\omega s k_L A, \tag{8}$$

$$\sigma_{xz} = \mu(\partial_z u_x + \partial_x u_z) = -2\mu\,\omega^2\,s\,p\,k_L A, \tag{9}$$

$$\sigma_{zz} = \lambda\nabla\cdot\mathbf{u} + 2\mu\,\partial_z u_z = -\omega^2\Big(\tfrac{\lambda}{V_L^2} + 2\mu k_L^2\Big)A = -\omega^2(\rho - 2\mu p^2)A, \tag{10}$$

where $\nabla\cdot\mathbf{u} = \nabla^2\phi = -\omega^2 A/V_L^2$ and $\lambda/V_L^2 + 2\mu k_L^2 = (\lambda+2\mu)/V_L^2 - 2\mu p^2 = \rho - 2\mu p^2$. For an SV wave, $\nabla\cdot\mathbf{u} = 0$ and

$$u_x = -\mathrm{i}\omega s k_T B, \qquad u_z = \mathrm{i}\omega p\,B, \tag{11}$$

$$\sigma_{xz} = \mu\omega^2(k_T^2 - p^2)B = \omega^2(\rho - 2\mu p^2)B, \tag{12}$$

$$\sigma_{zz} = 2\mu\,\partial_z u_z = -2\mu\,\omega^2\,s\,p\,k_T B, \tag{13}$$

where $\mu(k_T^2 - p^2) = \rho - 2\mu p^2$ was used. The combination

$$D \equiv \rho - 2\mu p^2 = \rho\,(1 - 2V_T^2 p^2) \tag{14}$$

appears everywhere; for a transverse wave of angle θ it equals $\rho\cos 2\theta$.

All fields scale as ωA (displacement) or $\omega^2 A$ (stress). Defining the *reduced* fields $\tilde{u} = u/\omega$ and $\tilde{\sigma} = \sigma/\omega^2$, the frequency drops out of every equation except through the bond stiffness (Sec. 7). Table 2 lists the reduced field vectors $(\tilde{u}_x, \tilde{u}_z, \tilde{\sigma}_{xz}, \tilde{\sigma}_{zz})$ per unit potential amplitude.

Table 2: Reduced field vectors per unit potential amplitude. $D = \rho - 2\mu p^2$ is evaluated with the density and shear modulus of the medium containing the wave.

wave	$\tilde{u}_x$	$\tilde{u}_z$	$\tilde{\sigma}_{xz}$	$\tilde{\sigma}_{zz}$
L	$\mathrm{i}p$	$\mathrm{i}sk_L$	$-2\mu s\,p\,k_L$	$-D$
SV	$-\mathrm{i}sk_T$	$\mathrm{i}p$	D	$-2\mu s\,p\,k_T$

6 Wave content of the problem

An incident wave in the crystal (type L or SV, $s = +1$) generates a reflected L and a reflected SV ($s = -1$) in the crystal, and a transmitted L and a transmitted SV ($s = +1$) in the metal. Their potential amplitudes are

$$\mathbf{x} = (A_r,\, B_r,\, A_t,\, B_t)^{\mathsf{T}}. \tag{15}$$

The incident potential is normalised to unit displacement amplitude. For a propagating wave $|\tilde{u}|^2 = (p^2 + k_j^2)|A|^2 = |A|^2/V^2$, so $A_{\mathrm{inc}} = V_{\mathrm{inc}}$, and the amplitude coefficients are

$$\Gamma_{cL} = \frac{A_r}{V_{L,c}}, \quad \Gamma_{cT} = \frac{B_r}{V_{T,c}}, \quad \Gamma_{mL} = \frac{A_t}{V_{L,m}}, \quad \Gamma_{mT} = \frac{B_t}{V_{T,m}}. \tag{16}$$

For propagating waves, $|\Gamma|$ is the displacement amplitude relative to the incident one; for an evanescent wave, Γ is only a rescaled potential amplitude.

7 Boundary conditions

The interface is a layer of distributed springs with tangential stiffness $K_\parallel$ and normal stiffness $K_\perp$ (stress per unit relative displacement). The conditions are

$$\text{continuity of traction:} \qquad \sigma_{xz}^c = \sigma_{xz}^m, \qquad \sigma_{zz}^c = \sigma_{zz}^m, \tag{17}$$

$$\text{spring law:} \qquad \sigma_{xz}^c = K_\parallel\,(u_x^m - u_x^c), \qquad \sigma_{zz}^c = K_\perp\,(u_z^m - u_z^c), \tag{18}$$

where the superscript denotes the total field (incident plus reflected for c; transmitted for m) at $z = 0$. The limits are

$$K \to \infty:\ u^c = u^m \ \text{(welded)}, \qquad K \to 0:\ \sigma^c = 0 \ \text{(free surface; the metal decouples)}. \tag{19}$$

Dividing Eq. (18) by ω^2 and using $u = \omega\tilde{u}$ gives $\tilde{\sigma} = (K/\omega)\,\Delta\tilde{u}$. The coupling is expressed through the dimensionless parameters

$$\alpha_\parallel = \frac{K_\parallel}{\rho_c V_{T,c}\,\omega}, \qquad \alpha_\perp = \frac{K_\perp}{\rho_c V_{L,c}\,\omega}, \tag{20}$$

so that $\alpha \propto 1/\omega$ for a frequency-independent stiffness. The reduced stiffnesses are

$$\tilde{K}_\parallel = \frac{K_\parallel}{\omega} = \alpha_\parallel\,\rho_c V_{T,c} = \alpha_\parallel \frac{\mu_c}{V_{T,c}}, \qquad \tilde{K}_\perp = \frac{K_\perp}{\omega} = \alpha_\perp\,\rho_c V_{L,c} = \alpha_\perp \frac{\lambda_c + 2\mu_c}{V_{L,c}}. \tag{21}$$

8 The 4×4 linear system

Write $\tilde\sigma^{w}$ and $\tilde u^{w}$ for the entries of Table 2 of wave $w \in \{rL, rT, tL, tT, \mathrm{inc}\}$. Eq. (17) gives

$$\tilde\sigma^{rL}_{xz} A_r + \tilde\sigma^{rT}_{xz} B_r - \tilde\sigma^{tL}_{xz} A_t - \tilde\sigma^{tT}_{xz} B_t = -\tilde\sigma^{\mathrm{inc}}_{xz}, \tag{22}$$

$$\tilde\sigma^{rL}_{zz} A_r + \tilde\sigma^{rT}_{zz} B_r - \tilde\sigma^{tL}_{zz} A_t - \tilde\sigma^{tT}_{zz} B_t = -\tilde\sigma^{\mathrm{inc}}_{zz}. \tag{23}$$

Substituting $\tilde\sigma^{c} = \tilde\sigma^{\mathrm{inc}} + \tilde\sigma^{r}$ and $\tilde u^{c} = \tilde u^{\mathrm{inc}} + \tilde u^{r}$ into Eq. (18) gives, for the tangential component,

$$\big(\tilde\sigma^{rL}_{xz} + \tilde K_\parallel \tilde u^{rL}_{x}\big) A_r + \big(\tilde\sigma^{rT}_{xz} + \tilde K_\parallel \tilde u^{rT}_{x}\big) B_r - \tilde K_\parallel \tilde u^{tL}_{x} A_t - \tilde K_\parallel \tilde u^{tT}_{x} B_t = -\big(\tilde\sigma^{\mathrm{inc}}_{xz} + \tilde K_\parallel \tilde u^{\mathrm{inc}}_{x}\big), \tag{24}$$

and the normal component is identical with $(x, xz, \parallel) \to (z, zz, \perp)$. In matrix form $M\mathbf{x} = \mathbf{b}$ with

$$M = \begin{pmatrix} \tilde\sigma^{rL}_{xz} & \tilde\sigma^{rT}_{xz} & -\tilde\sigma^{tL}_{xz} & -\tilde\sigma^{tT}_{xz} \\ \tilde\sigma^{rL}_{zz} & \tilde\sigma^{rT}_{zz} & -\tilde\sigma^{tL}_{zz} & -\tilde\sigma^{tT}_{zz} \\ \tilde\sigma^{rL}_{xz} + \tilde K_\parallel \tilde u^{rL}_{x} & \tilde\sigma^{rT}_{xz} + \tilde K_\parallel \tilde u^{rT}_{x} & -\tilde K_\parallel \tilde u^{tL}_{x} & -\tilde K_\parallel \tilde u^{tT}_{x} \\ \tilde\sigma^{rL}_{zz} + \tilde K_\perp \tilde u^{rL}_{z} & \tilde\sigma^{rT}_{zz} + \tilde K_\perp \tilde u^{rT}_{z} & -\tilde K_\perp \tilde u^{tL}_{z} & -\tilde K_\perp \tilde u^{tT}_{z} \end{pmatrix}, \quad \mathbf{b} = -\begin{pmatrix} \tilde\sigma^{\mathrm{inc}}_{xz} \\ \tilde\sigma^{\mathrm{inc}}_{zz} \\ \tilde\sigma^{\mathrm{inc}}_{xz} + \tilde K_\parallel \tilde u^{\mathrm{inc}}_{x} \\ \tilde\sigma^{\mathrm{inc}}_{zz} + \tilde K_\perp \tilde u^{\mathrm{inc}}_{z} \end{pmatrix}. \tag{25}$$

In the welded limit rows 3 and 4 are divided by $\tilde K$ and $\tilde K \to \infty$, giving $\tilde u^{rL}_{x} A_r + \tilde u^{rT}_{x} B_r - \tilde u^{tL}_{x} A_t - \tilde u^{tT}_{x} B_t = -\tilde u^{\mathrm{inc}}_{x}$ and the same for z. The system has no useful closed form for general θ_i; it is solved numerically as a complex 4×4 system at each angle, and the amplitudes follow from Eq. (16). Closed forms exist only in the special cases of Sec. 11.

9 SH waves

For polarisation along y, $u_y = C\, e^{\mathrm{i}\omega(px + s k_T z - t)}$ and the only relevant stress is $\sigma_{yz} = \mu\, \partial_z u_y = \mathrm{i}\omega\mu\, s\, k_T\, u_y$. The two conditions at $z = 0$ are continuity of σ_{yz} and the spring law $\sigma^{c}_{yz} = K_{SH}(u^{m}_{y} - u^{c}_{y})$ with $\tilde K_{SH} = \alpha_{SH}\mu_c / V_{T,c}$. For unit incident amplitude, reflected amplitude R and transmitted amplitude T_r,

$$a(1 - R) = b\, T_r, \tag{26}$$

$$\mathrm{i}a(1 - R) = \tilde K_{SH}\,(T_r - 1 - R), \tag{27}$$

with $a = \mu_c k_{T,c}$ and $b = \mu_m k_{T,m}$. The solution is

$$T_r = \frac{2}{1 + b/a - \mathrm{i}\, b/\tilde K_{SH}}, \qquad R = 1 - \frac{b}{a} T_r. \tag{28}$$

The model sets $\alpha_{SH} = \alpha_\parallel$, because SH motion is purely tangential (an assumption of this model).

10 Energy flux and power coefficients

For a propagating plane wave with displacement amplitude a, the time-averaged energy density is $\tfrac{1}{2}\rho\omega^2|a|^2$. It travels at velocity V along the propagation direction, so its z-flux is $\langle P_z \rangle = \tfrac{1}{2}\rho\,\omega^2|a|^2\, V \cos\theta$, with $\cos\theta = Vk$ from Eq. (7). The fraction of the incident power carried by the outgoing wave j is therefore

$$W_j = |\Gamma_j|^2\, \frac{\rho_j V_j \cos\theta_j}{\rho_{\mathrm{inc}} V_{\mathrm{inc}} \cos\theta_{\mathrm{inc}}} \quad \text{(propagating)}, \qquad W_j = 0 \quad \text{(evanescent)}. \tag{29}$$

Cross terms between different modes do not contribute, by modal power orthogonality in a lossless medium (Auld, *Acoustic Fields and Waves in Solids*, Vol. II). The transmission probability is $W_{\mathrm{trans}} = W_{mL} + W_{mT}$ and the reflection probability is $W_{\mathrm{refl}} = W_{cL} + W_{cT}$.

Energy conservation. The power flowing in $+z$ is $-\frac{1}{2}\,\mathrm{Re}[\sigma_{iz}\,v_i^*]$ with $v = -\mathrm{i}\omega u$. Traction is continuous, so with $\sigma \equiv \sigma^c = \sigma^m$,

$$P_z^c - P_z^m = -\tfrac{1}{2}\,\mathrm{Re}\big[\,\mathrm{i}\omega\,\sigma_{iz}\,(u_i^c - u_i^m)^*\big] = \tfrac{1}{2}\,\mathrm{Re}\big[\mathrm{i}\omega\,K\,|\Delta u|^2\big] = 0 \tag{30}$$

for real K (using $\sigma = K\,\Delta u$, $\Delta u = u^m - u^c$, summed over $i = x, z$). A purely elastic spring stores energy but dissipates none, so $W_{cL} + W_{cT} + W_{mL} + W_{mT} = 1$ at every angle. This was verified numerically to 10^{-8} for the welded and spring cases and for SH.

11 Closed-form checks

Normal incidence. With $p = 0$ the L and SV problems decouple. For one wave type with impedances $Z_c = \rho_c V_c$ and $Z_m = \rho_m V_m$, write $u_c = e^{\mathrm{i}k_c z} + Re^{-\mathrm{i}k_c z}$ and $u_m = Te^{\mathrm{i}k_m z}$. Then $\sigma_c = \mathrm{i}\omega Z_c(1 - R)$ and $\sigma_m = \mathrm{i}\omega Z_m T$. Continuity of traction gives $Z_c(1 - R) = Z_m T$, and the spring law with $K = \alpha Z_c \omega$ gives $\mathrm{i}Z_m T = \alpha Z_c(T - 1 - R)$. With $r = Z_m/Z_c$,

$$T = \frac{2}{(1 + r) - \mathrm{i}\,r/\alpha}, \qquad W_{\mathrm{trans}} = r|T|^2 = \frac{4r}{(1 + r)^2 + r^2/\alpha^2}. \tag{31}$$

For $\alpha \to \infty$ this is the acoustic-mismatch result $4r/(1 + r)^2$. For sapphire/Ta (transverse impedances $Z_c = 2.57 \times 10^7$ and $Z_m = 3.43 \times 10^7$ kg m^{-2}s^{-1}, $r = 1.33$) and $\alpha = 1$ it gives 0.739, which is the left end of the orange curve in Fig. 1(d).

Free surface ($K \to 0$). The metal decouples and rows 1 and 2 reduce to $\sigma_{xz}^c = \sigma_{zz}^c = 0$. With an incident SV wave $B_{\mathrm{inc}} = V_{T,c}$ and $D = \rho_c \cos 2\theta$,

$$2\mu p\,k_L\,A_r + D\,B_r = -D\,B_{\mathrm{inc}}, \tag{32}$$

$$-D\,A_r + 2\mu p\,k_T\,B_r = 2\mu p\,k_T\,B_{\mathrm{inc}}. \tag{33}$$

The determinant is $\Delta = 4\mu^2 p^2 k_L k_T + D^2$ (the Rayleigh-type denominator), and

$$A_r = -\frac{4\mu p\,k_T\,D}{\Delta} B_{\mathrm{inc}}, \qquad B_r = \frac{4\mu^2 p^2 k_L k_T - D^2}{\Delta} B_{\mathrm{inc}}. \tag{34}$$

With $\kappa = V_L/V_T$, $\sin\theta_L = \kappa \sin\theta$, $\mu p k_T = \frac{1}{2}\rho \sin 2\theta$ and $\mu p k_L = \rho \sin 2\theta_L/(2\kappa^2)$,

$$\Gamma_{T\to L} = -\frac{\kappa \sin 4\theta}{\sin 2\theta\,\sin 2\theta_L + \kappa^2 \cos^2 2\theta}, \qquad \Gamma_{T\to T} = \frac{\sin 2\theta\,\sin 2\theta_L - \kappa^2 \cos^2 2\theta}{\sin 2\theta\,\sin 2\theta_L + \kappa^2 \cos^2 2\theta}. \tag{35}$$

12 Critical angle and the structure of Fig. 1(c,d)

For an incident SV wave, $p = \sin\theta/V_{T,c}$, and the reflected L wave in sapphire is propagating only if $p < 1/V_{L,c}$, that is

$$\theta < \theta_c = \arcsin\frac{V_{T,c}}{V_{L,c}} = 36.4^\circ. \tag{36}$$

At θ_c the reflected L wave grazes the surface ($\cos\theta_L = 0$, $\sin 2\theta_L = 0$); above θ_c it is evanescent and $\sin 2\theta_L$ becomes imaginary in Eq. (35). At θ_c the free-surface amplitude is finite:

$$|\Gamma_{T\to L}|^2_{\theta_c} = \left(\frac{\sin 4\theta_c}{\kappa \cos^2 2\theta_c}\right)^2 = 14.7. \tag{37}$$

The peak is a cusp from the square-root branch point of k_L, not a divergence; on a finite angular grid the plotted maximum is slightly lower because the grid does not land exactly on θ_c.

The quantity in panel (c) is a *displacement amplitude* squared, not a power, so it can exceed one without violating energy conservation: the associated power is $W_{cL} = |\Gamma_{T\to L}|^2\,\kappa\cos\theta_L/\cos\theta$ from Eq. (29), which goes to zero at θ_c. For SV incidence the metal waves are always propagating because sapphire is the fast medium: for Ta, $p_{\max} = 1/V_{T,c} = 0.155$ s/km is below $1/V_{L,m} = 0.239$ s/km and $1/V_{T,m} = 0.486$ s/km. The cusp in panel (d) is the transmission side of the same branch point.

13 Angular and Debye averages

For isotropic phonon incidence from the crystal, the number flux per solid angle is proportional to $\cos\theta$, so the flux-weighted transmission of a mode is

$$\langle W\rangle_\nu = \frac{\int_0^{\pi/2} W_\nu(\theta)\cos\theta\sin\theta\,d\theta}{\int_0^{\pi/2}\cos\theta\sin\theta\,d\theta} = 2\int_0^{\pi/2} W_\nu(\theta)\cos\theta\sin\theta\,d\theta, \tag{38}$$

with $\nu \in \{\mathrm{L},\mathrm{SV},\mathrm{SH}\}$ and W_ν the transmission probability for an incident wave of type ν (W_{trans} from Sec. 9 for SH). The average over the two transverse polarisations is $\langle W\rangle_T = \frac{1}{2}(\langle W\rangle_{SV} + \langle W\rangle_{SH})$. Weighting by the Debye density of states, $\propto V^{-3}$ per mode (one L and two T), gives

$$\langle W\rangle_{\mathrm{Debye}} = \frac{2V_{T,c}^{-3}\langle W\rangle_T + V_{L,c}^{-3}\langle W\rangle_L}{2V_{T,c}^{-3} + V_{L,c}^{-3}}. \tag{39}$$

The frequency enters only through $\alpha \propto 1/\omega$ (Eq. (20)). The relation between α and phonon energy in Fig. 2 ($\alpha = 1$ at 2.48 meV, K frequency independent) is an assumption, not a result of the model.

14 Figures

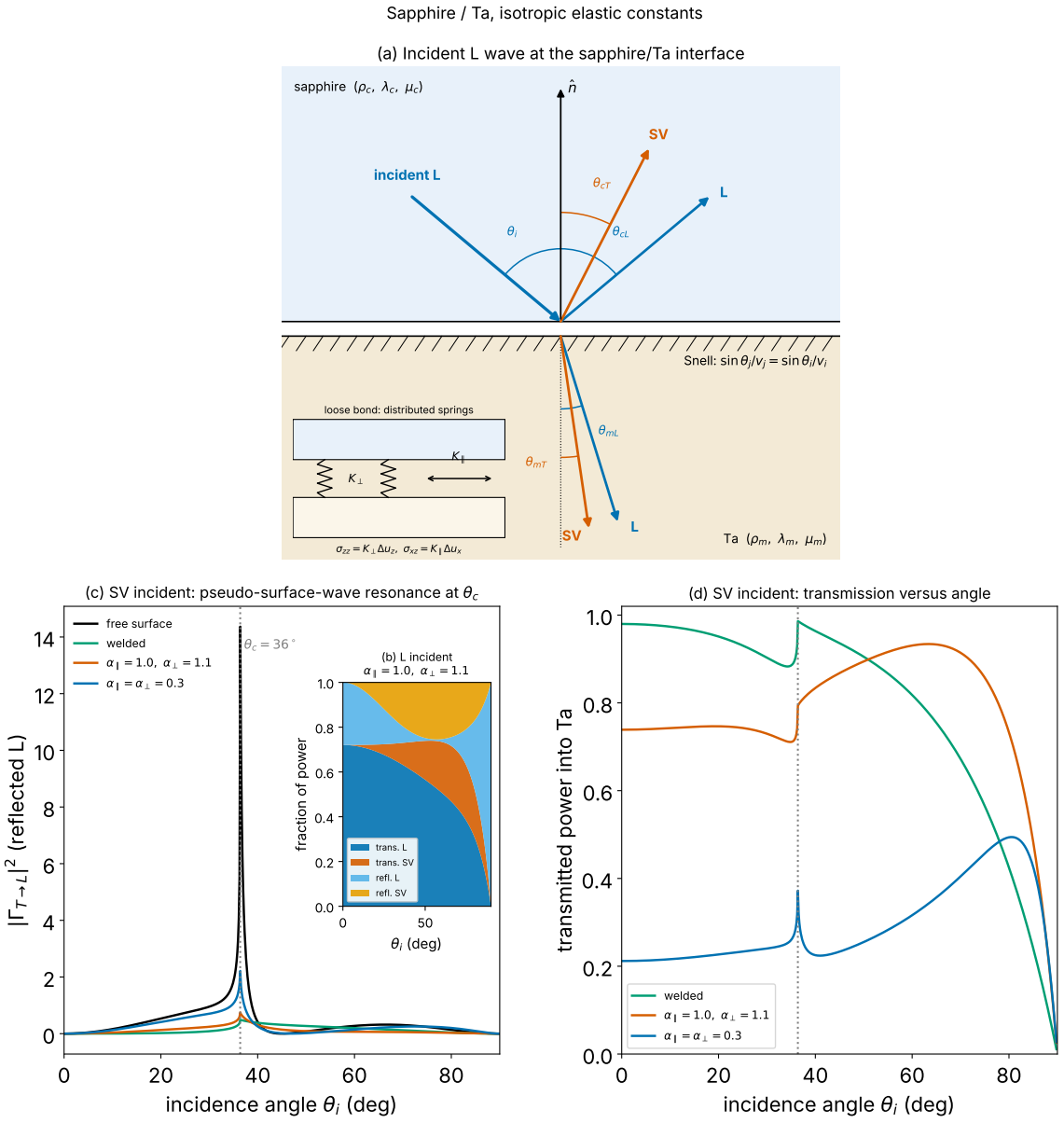


Figure 1: Sapphire/Ta interface: (a) ray diagram with the loose-bond boundary condition; (b) power partition of an incident L wave; (c) reflected-L amplitude for an incident SV wave; (d) power transmitted into Ta for an incident SV wave.

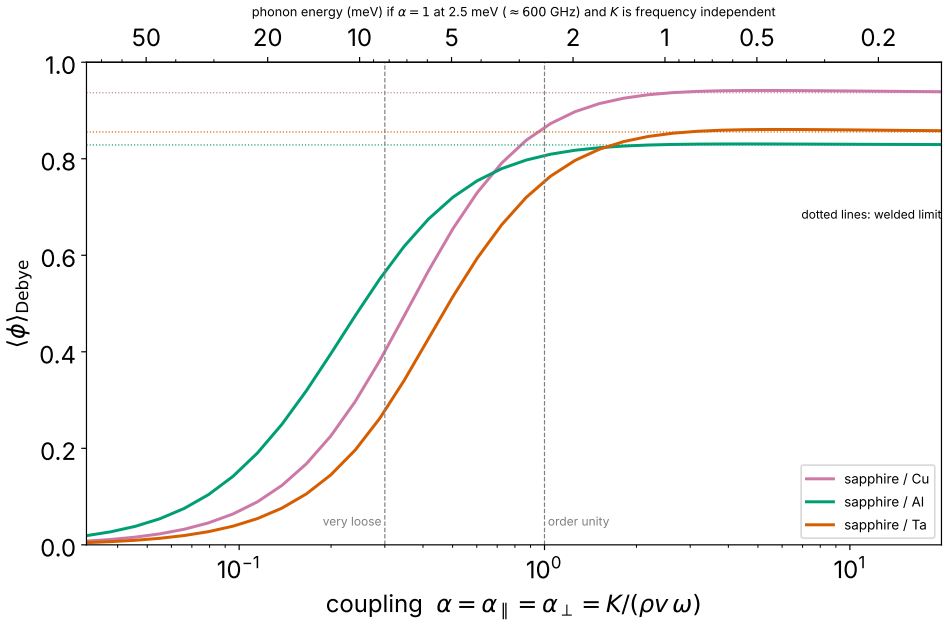


Figure 2: Debye- and flux-averaged transmission probability versus coupling $\alpha = \alpha_{\parallel} = \alpha_{\perp}$.

References

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